AnswerBot

AI-powered FAQ and Customer Support Automation

REST API platform for secure FAQ management, AI generation and semantic search

Node.js • Express.js • MongoDB • Mongoose • JWT • bcrypt • Google Gemini

1. Project Overview

- Robust and secure REST API platform for FAQ creation and customer support automation.
- Built using Node.js and Express.js with MongoDB as the primary data store.
- Mongoose provides data modeling and schema validation.
- JWT enables stateless authentication and route guarding.
- bcrypt protects passwords through hashing.
- Backend supports AI-powered FAQ generation, semantic search and context parsing.
- Centralized error handling improves operational reliability.

2. Problem Statement

Scenario: Priya, a customer support operations manager

Documentation

Difficulty generating and maintaining up-to-date documentation workflows.

Search

Limited visibility into semantic-search keywords and rapid query resolution.

Manual Work

Time-consuming manual drafting of complex support answers.

Permissions

Need for unified database permission layers for private documentation.

3. Proposed Solution

- Centralized and secure API wrapper separates backend business logic from frontend delivery.
- Verification controls manage content creation, category keywords, database persistence and AI text inference.
- Authenticated creators can generate, create, update and delete their own FAQ records.
- Public/consumer users can browse published FAQs and perform localized semantic queries.
- Schema validation maintains structural integrity, while security tokens protect modification access.
- Indexing improves real-time FAQ discoverability.

4. Software & Hardware Requirements

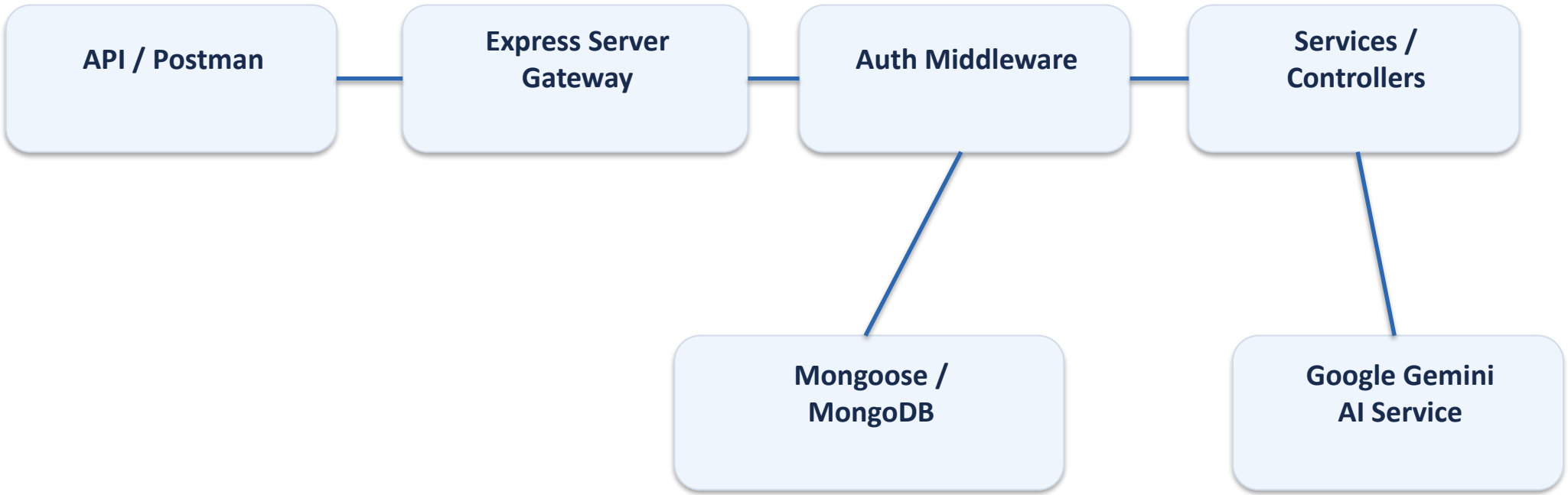
Software Requirements

- Windows 10/11, macOS or Linux
- Node.js v18+
- npm v9+
- Express.js v4/v5 ecosystem
- MongoDB / Mongoose ODM
- Postman / ThunderClient
- Visual Studio Code or similar IDE

Hardware Requirements

- Intel Core i5 8th Gen+ / AMD Ryzen 5 equivalent
- RAM: 8 GB minimum
- 16 GB recommended for concurrent MongoDB processing
- Storage: 1 GB available workspace

5. Project Architecture



Requests are filtered and authenticated before business logic reaches data services or AI integration.

6. Data Model / ER Structure

User Entity

_id (ObjectId)
name
email (unique)
password (hashed)
role
createdAt / updatedAt

FAQ Entity

_id
question
answer
category
createdBy → User
createdAt / updatedAt

AI Generation

topic
generatedQuestion
generatedAnswer
generatedCategory
generatedAt

Category

categoryName
description
totalFAQs

Mappings: User → FAQ (1:M) • Category → FAQ (1:M) • User → AI-generated FAQ (1:M) • FAQ → Search Results

7. Key Features

- ✓ JWT Authentication & Authorization
- ✓ AI FAQ Generation
- ✓ Semantic FAQ Search
- ✓ Complete CRUD Operations
- ✓ Category Management
- ✓ Password Encryption
- ✓ Centralized Error Handling
- ✓ Mongoose Schema Validation
- ✓ RESTful API Architecture
- ✓ MongoDB Integration

8. User Roles & Responsibilities

Admin

- Manage registered users
- View/update/delete any FAQ
- Monitor AI-generated content
- Manage FAQ categories
- Access complete system functionality

Content Creator

- Create new FAQs
- Edit personal FAQs
- Delete owned FAQs
- Generate FAQs using AI
- Organize FAQs by category

Authenticated User

- View public FAQs
- Search using keywords
- Generate AI-assisted answers
- Access personal profile

Public User

- View published FAQs
- Search FAQs
- No create/edit/delete access
- No protected endpoint access

9. MVC Architecture Pattern

Model Layer

Data blueprint

- Mongoose schemas
- Field rules and defaults
- Validation before persistence
- Maps data to MongoDB collections

Controller Layer

Execution matrix

- Captures requests
- Checks constraints
- Delegates to models/services
- Returns standard JSON responses

View / Routing Layer

API routing interface

- Headless backend
- No standard UI templates
- Maps network paths to controller handlers

10. Project Setup & Configuration

- Create the project folder and a src subfolder.
- Open the project folder in Visual Studio Code.
- Initialize the Node.js project using npm init -y.
- Install core dependencies: express, mongoose, dotenv, cors, jsonwebtoken, bcrypt and @google/genai.
- Organize the application into server, app, middleware, models and routes modules.
- Run the development server with npm run dev.

```
npm init -y     |     npm install express mongoose dotenv cors jsonwebtoken bcrypt @google/genai     |     npm
                                                                                                    run dev
```

11. Backend & Database Development

Server

src/server.js
- Loads environment variables
- Starts database listeners
- Binds server to active port

Express & Security

src/app.js
authMiddleware.js
errorMiddleware.js
- JSON parsing
- CORS
- JWT validation
- Centralized errors

Models

User.js
- Profile validation
- Password hashing

FAQ.js
- FAQ properties
- Creator reference

Routes

authRoutes.js
faqRoutes.js
aiRoutes.js

Public, protected and AI automation endpoints

12. API Testing & Execution

1. Register

POST /api/auth/register
Public access

Creates a user profile with name, email and password.

2. Login

POST /api/auth/login
Public access

Verifies profile credentials and establishes authenticated access.

3. Create FAQ

POST /api/faqs
Private
Bearer JWT required

Stores question, answer and category.

4. Search / AI

GET /api/faqs/search?q=configure

POST /api/ai/generate-faq
Private + Bearer JWT

13. Conclusion

- AnswerBot provides a centralized backend for FAQ creation, management and customer-support automation.
- Security is supported through JWT authentication, bcrypt password hashing and protected routes.
- MongoDB with Mongoose provides structured persistence and schema validation.
- Google Gemini integration supports AI-assisted FAQ generation and topic workflows.
- Role-based access and RESTful endpoints organize public, authenticated and administrative operations.
- API testing through Postman or ThunderClient verifies the main application endpoints.

Project document also provides API testing video and demo links.